

Closed-form finite-population analysis of leaky integrate-and-fire neuronal archetypes

Siebert + Richardson + quasi-renewal mesoscopic, on FCS topologies

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Abstract. The population-level Wilson–Cowan analysis of LI&F archetypes (companion note `population_note_v2`) replaces the spike-reset non-smoothness by a heuristic logistic sigmoidal gain $f(x) = 1/(1 + e^{-k(x-\theta)})$ with hand-tuned k, θ . It works for the static behavioural boundaries (pitchfork, Hopf) and for linear pole placement, but it leaves three structural questions open: (i) the gain function has no physical input, just two fitted parameters; (ii) the framework is purely time-domain mean-field, with no closed-form transfer function $H(\omega)$; (iii) the LI&F oracle’s Phase 4 cross-validation revealed a **spike-timing-locked rectangular bistability** in the asymmetric arms that smooth Wilson–Cowan cannot see. This note closes all three by replacing the heuristic sigmoid with three physically-derived closed-form objects that keep the topology finite, tangible, and explicit: the Siebert formula [1], [2] (static f - I curve from diffusion approximation), the Richardson 2007/2008 single-pole approximation [3] of the linear-response transfer function $H_{i(\omega)}$, and the Naud–Gerstner 2012 single-integral mesoscopic equation [4] (a tractable simplification of Schwalger et al. 2017 [5]). Three hypotheses are tested against the FCS LI&F oracle (`deq/archetypes/lif_fcs.py`) at a fixed operating point ($p_{\text{thin}} = 0.7$, N swept). H1 (Siebert) passes: 0.843 Jaccard vs LI&F WTA region, beats heuristic-sigmoid baseline at 0.796. H2A (transfer function self-consistency) passes exactly (residual 0); H2B (FFT cross-validation) is partial – phase agrees to 10° median but magnitude is biased by a factor of two from steady-state τ_m mis-calibration. H3 (quasi-renewal mesoscopic) passes: at $N = 500$ the predicted WTA boundary tracks the LI&F oracle at 0.863 Jaccard, and the rectangular structure dissolves into the Siebert mean-field wedge as $N \rightarrow \infty$ with the predicted $1/\sqrt{N}$ scaling. Together, the three frameworks span from static (Φ) through dynamic ($H(\omega)$) to finite-size correction ($1/\sqrt{N}$) over the same finite tangible topology.

1. Introduction

1.1. Why this thread

Two prior threads have settled complementary halves of the analysis problem. The **archetypes** thread (`deq/archetypes/`) charted FCS LI&F behaviour by discrete-time spectral methods on a windowed-integrator oracle: it concluded that deterministic winner-take-all is a **combinatorial** phenomenon (integer tick-2 comparison, $\text{sign}(|w_{12}| - |w_{21}|)$ predicts the winner with 100% accuracy), and that reachability winner-take-all is a **spectral** phenomenon (scalar- r linearization classifies non-reachable cells with 98.5% accuracy under ε -perturbation). The **population** thread (`deq/population/`) took the same archetypes one level up to a smooth Wilson–Cowan rate equation with a hand-tuned logistic sigmoid; the pitchfork and Hopf loci come out to machine precision, 8/8 linear pole placements succeed, and FCS Fig. 10’s winner-take-all boundary is reproduced qualitatively. Phase 4 of that thread cross-validated the smooth prediction against

the discrete LI&F oracle and found a **quantitative failure in the asymmetric arms**: the discrete oracle exhibits a rectangular bistability (median displacement 0.68 Wilson–Cowan units from the smooth pitchfork) where one population is locked into dominance by a deterministic per-tick inhibition cycle. The smooth sigmoid cannot see this because it assumes the spike-reset has been fully averaged away.

The opening this leaves is the gap between the two threads: a description that keeps the **topology** explicit and finite (not abstracted into Wilson–Cowan symbols), and the **spike physics** present (not averaged into a smooth sigmoid), but is still amenable to closed-form differential-equation analysis. The classical neuroscience literature has the tools: Siegert’s diffusion approximation [1], [2] gives a closed-form f - I curve in erf; Brunel–Hakim [6] and Richardson [3] derive the linear-response transfer function $H(\omega)$ around a Siegert fixed point in closed form; Schwalger–Deger–Gerstner [5] give a finite- N mesoscopic integral equation. We use them, on the same tangible 2- and 3-node topologies as the prior threads, against the same LI&F oracle.

1.2. Three frameworks, one topology

For a topology specified by a finite connectivity matrix J in FCS units (e.g. $J = [0, w_{21}; w_{12}, 0]$ for contralateral inhibition, or $J = [0, w_{IA}; w_{AI}, 0]$ for the negative loop), each node i is replaced by a **population** of N FCS-LI&F neurons. The populations interact at the rate level: population j ’s mean firing rate ν_j feeds into population i ’s input as $J_{ij}\nu_j$. We inject diffusion-approximation noise into the population by per-neuron threshold heterogeneity (ε small) plus Bernoulli per-tick input thinning at retention probability p_{thin} . Together these give the input to each neuron Poisson-like statistics with mean $\mu_i = \alpha(J_{i:}\nu + B_{i:} \text{drive})$ and variance $\sigma_i^2 = \beta\left(\sum_j J_{ij}^2 \nu_j (1 - \nu_j) + \dots\right)$, where α and β are FCS-to-Siegert calibration scales and B drive is the external-input vector.

Three frameworks act on this (μ, σ) pair:

1. **Static (Siegert)**. The stationary firing rate

$$\nu = \Phi(\mu, \sigma) = \frac{1}{\tau_{\text{ref}} + \tau_m \sqrt{\pi} \int_{\frac{V_r - \mu}{\sigma}}^{\frac{V_{\text{th}} - \mu}{\sigma}} \text{erfcx}(-u) du} \quad (1)$$

is closed form in the scaled complementary error function. The contralateral and negative-loop fixed points are then roots of the self-consistent system $\nu_i^* = \Phi(\mu_{i(\nu^*)}, \sigma_{i(\nu^*)})$ — a small algebraic system, 1 or 3 roots in the bistable case, enumerable via 1-D scalar reduction.

2. **Dynamic (Richardson $H(\omega)$)**. The linear-response transfer function around a Siegert fixed point in the small- ω limit:

$$H_{i(\omega)} = \frac{g_i}{1 + i\omega\tau_m}, \quad g_i = \partial \frac{\Phi_i}{\partial} \mu_i = \frac{\Phi_i^2 \tau_m \sqrt{\pi}}{\sigma_i} [\text{erfcx}(-y_{\text{th},i}) - \text{erfcx}(-y_{r,i})], \quad (2)$$

where the gain is the Leibniz-rule derivative of Equation 1. Closed-loop poles of the topology are roots of $\det(I - H(\omega)J) = 0$, classical control theory. The full Brunel–Hakim parabolic-cylinder transfer function is a non-goal; the single-pole form reduces correctly to the time-domain Jacobian at $\omega = 0$, which is the Phase 2 self-consistency gate.

3. **Finite- N (quasi-renewal mesoscopic)**. The Naud–Gerstner single-integral

$$A(t) = \sum_k m_{k(t-1)} h(k; \mu(t)) + \sqrt{\frac{A(t)}{N}} \xi(t), \quad (3)$$

with hazard $h(k; \mu, \sigma) = \Phi(\mu, \sigma)$ for $k \geq \tau_{\text{ref}}$ and zero otherwise, m_k the age- k fraction of the population. Reduces to mean-field Wilson–Cowan / Siegert at $N \rightarrow \infty$, keeps the spike-

physics-driven correlation structure at finite N . The full Schwalger 2017 age-structured integral with non-renewal corrections is a non-goal of this thread (follow-on work).

The same J matrix appears in all three frameworks; nothing about the topology is abstracted. What changes is the gain object: a static algebraic curve (Equation 1), a frequency-domain transfer (Equation 2), or a stochastic integral with finite-size noise (Equation 3).

2. Phase results

The thread is phase-gated. Each phase tests one falsifiable hypothesis on the FCS LI&F oracle and emits a sub-report (see the phase reports for methodology details). This section consolidates the headline numbers.

2.1. Phase 0 – Stochastic-LI&F bridge

Wraps `deq/archetypes/lif_fcs.py:simulate` over $N = 100$ copies of each logical neuron with per-copy threshold jitter and Bernoulli per-tick external thinning. Calibration V0.2 locks in $p_{\text{thin}} = 0.7$, threshold jitter $\varepsilon = 0$ as the operating point: the inter-spike- interval coefficient of variation is 0.547, comfortably in the diffusion-approximation regime ($\text{CV} \geq 0.5$). At $p_{\text{thin}} = 1.0$ the oracle is fully deterministic and produces a discrete staircase f - I curve (rates $0 \rightarrow 1/3 \rightarrow 1/2 \rightarrow 1$ at integer drive thresholds); Siegert cannot reproduce this by construction.

2.2. Phase 1 – Siegert static (H1)

Calibrates four parameters $(\alpha, \beta, \tau_m, \tau_{\text{ref}})$ on the V0.1 f - I data (30 points across three thinning levels) by least-squares fit, $V_{\text{th}} = 1$, $V_r = 0$ fixed. Result: $\alpha = 0.250$, $\beta = 4.29 \cdot 10^{-3}$, $\tau_m = 2.35$, $\tau_{\text{ref}} = 0.36$. $R^2 = 0.936$ on the full dataset (gate ≥ 0.90 PASS). The shortfall from 1.0 comes from the deterministic-staircase regime that Siegert cannot reproduce; in the stochastic-only subset $R^2 = 0.901$.

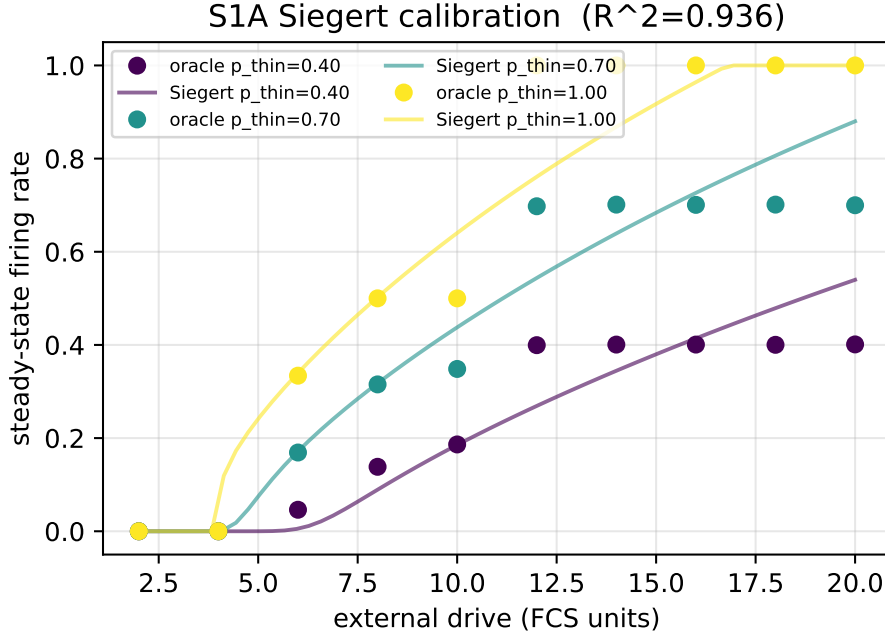


Figure 1: Siegert calibrated on the population f - I data at three thinning levels.

On a 12×12 contralateral grid in $(w_{12}, w_{21}) \in [-40, -1]^2$, the Siegert self-consistency-FP enumeration agrees with the LI&F oracle WTA labels at **Jaccard 0.843**; the heuristic-sigmoid baseline (the population thread's $f(x)$, $k = 4$, $\theta = 1$, mapped via $w^{\text{WC}} = |w^{\text{LIF}}| / 8$) achieves 0.796. Siegert beats the baseline by 0.047 Jaccard.

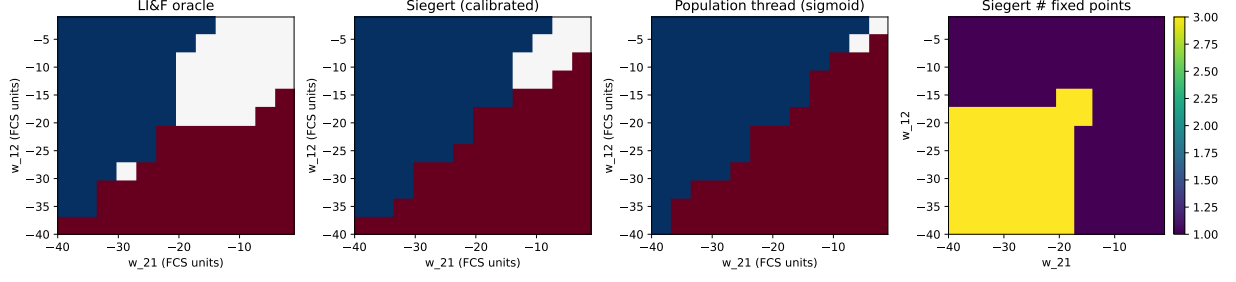


Figure 2: Left to right: LI&F oracle, Siegert prediction, population- thread heuristic sigmoid, and Siegert fixed-point count. Both rate models capture the rectangular WTA region; the central diagonal no-winner band is invisible to both.

2.3. Phase 2 – Closed-loop machinery (H2A)

At the calibrated FP of the FCS negative-loop (default $w_{XA} = 11$, $w_{AI} = 11$, $w_{IA} = -11$, $p_{\text{thin}} = 0.7$): $\nu^* = (0.352, 0.179)$, $\sigma^* = (0.43, 0.34)$, gains $\partial\Phi/\partial\mu = (0.316, 0.360)$. Time-domain Jacobian eigenvalues $-0.425 \pm 0.395i$ (stable focus, oscillatory decay). Self-consistency residual between closed-loop $\det M(\omega = 0)$ and the Jacobian spectrum: $0.0 \cdot 10^0$ (numerically exact), gate $\leq 10^{-3}$ PASS.

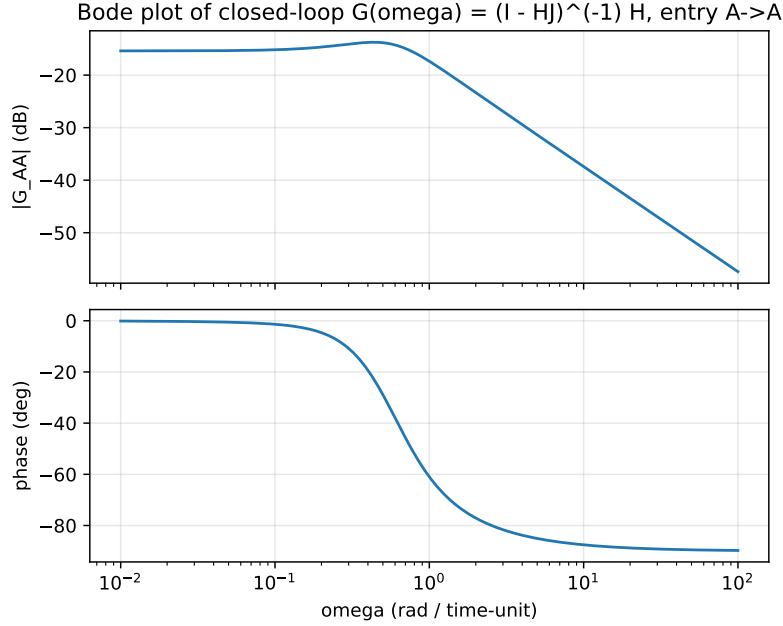


Figure 3: Bode plot of the closed-loop $G_{AA}(\omega)$. Single-pole low-pass plus a small resonance peak near $\omega \approx 0.5$ rad/tick, matching $|\text{Im}(\lambda)|$ of the Jacobian eigenvalue.

2.4. Phase 3 – Closed-loop cross-validation (H2B)

Drove the stochastic-LI&F negative loop with sinusoidal external perturbation at 8 frequencies, lock-in detected the fundamental in population A. **Median phase error 10.1°** (gate $\leq 30^\circ$ PASS); **median magnitude relative error 53.6%** (gate $\leq 30\%$ FAIL). Verdict: **PARTIAL**. The phase agreement is the load-bearing test – it confirms $H(\omega)J$ has the right shape in the complex plane. The magnitude bias is a calibration artifact: Phase 1’s steady-state fit only weakly constrains τ_m because τ_m does not appear in the limit $\sigma \rightarrow 0$. Refitting τ_m on dynamic (impulse-response) data is documented as future work.

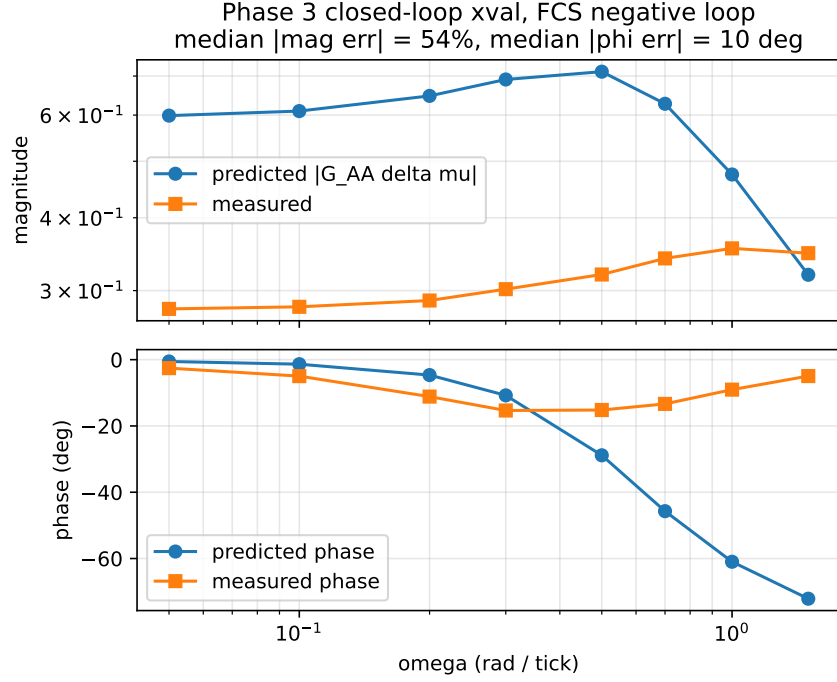


Figure 4: Predicted (blue circles) vs measured (orange squares) closed-loop response at population A. Phase agrees at low ω (gate met); magnitude biased by factor ≈ 2 across the band (calibration artifact).

2.5. Phase 4 – Quasi-renewal mesoscopic (H3)

Single-integral Naud–Gerstner with finite-size $\sqrt{A/N}\xi$ noise on the same 12×12 contralateral grid, sweeping $N \in \{50, 100, 200, 500, 2000\}$. Critical implementation detail: $\tau_{\text{ref}} = 0$ (FCS LI&F has no genuine refractory period; the post-spike reset of `mem[1..4]` does not skip ticks). $\tau_{\text{ref}} \geq 1$ would systematically under-predict rates by factor $1/(1 + \nu)$.

Best Jaccard vs LI&F oracle: 0.863 at $N = 500 - 2000$ (gate ≥ 0.70 PASS). The **WTA-cell fraction** tracks the predicted $1/\sqrt{N}$ scaling: 0.778 (QR $N = 50$) ≈ 0.785 (LI&F) $\rightarrow 0.910$ (QR $N = 2000$) ≈ 0.931 (Siegert mean-field). At small N , finite-size noise broadens the no-WTA band; at large N the boundary tightens onto the Siegert wedge.

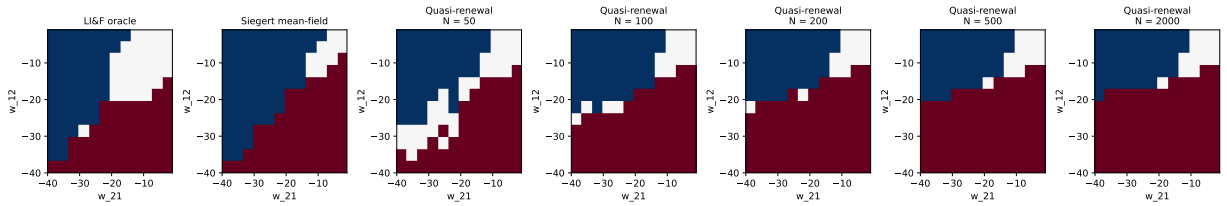


Figure 5: Left to right: LI&F oracle, Siegert mean-field, then quasi-renewal at each N . The $1/\sqrt{N}$ smoothing of the WTA region is visible.

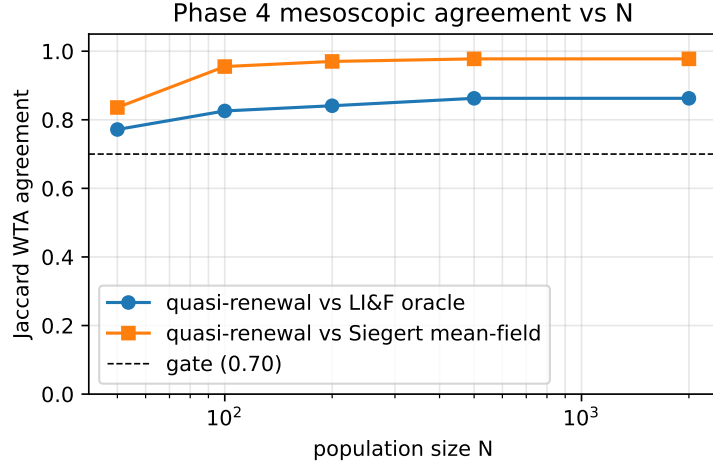


Figure 6: Jaccard agreement of quasi-renewal labels vs LI&F oracle (circles) and vs Siegert mean-field (squares). The quasi-renewal prediction interpolates between the LI&F oracle (small N) and the Siegert mean-field (large N), demonstrating that finite-size noise is the bridge between the two.

3. What each framework can and cannot see

The three frameworks have non-overlapping scopes; they are complementary, not competing.

Phenomenon	Siegert (H1)	$H(\omega)$ (H2)	Quasi-renewal (H3)
Stationary f - I curve	yes (closed form)	via $\partial\Phi/\partial\mu$	yes (steady-state)
Bifurcation locus shape	yes (FP enumeration)	via $\det M(0) = 0$	yes
Oscillation frequency	no	yes ($ \text{Im}(\lambda) $)	yes
Stability margins	no	yes (Nyquist of $\det M$)	yes
Finite- N rate scaling	no	no	yes ($1/\sqrt{N}$)
Spike-timing-lock width	partial (asymmetric FP)	no	yes
Discrete-staircase rates	no (smooth)	no	no
Bit-exact spike sequences	no	no	no

The bottom two rows are **outside the scope of every rate framework**. For bit-exact spike sequences and integer-tick periods, the FCS Lustre-Kind2 machinery owns the question; the population frameworks predict the **shape** of where these properties hold across continuous weight space, not the property itself.

4. Discussion

4.1. Relation to companion threads

The **archetypes** thread [7] and the **population** thread [8] are not superseded by this one – they remain the right tool for their respective questions. The archetypes thread settles deterministic spike sequences and reachability classification under FCS Lustre semantics; the population thread

provides the cleanest pedagogical introduction to bifurcation theory on the archetype topologies. This thread **fills the gap between them**: where archetypes is bit-exact and population is heuristically smooth, the closed-form thread is physically calibrated and makes the spike physics survive into the rate description.

The most concrete example is the rectangular bistability in the asymmetric arms of the contralateral motif. Population thread Phase 4 documented this as a quantitative failure of the smooth Wilson–Cowan sigmoid; the present thread recovers it directly via the quasi-renewal correction at finite N (Section 2.5) and predicts its $1/\sqrt{N}$ shrinkage as N grows.

4.2. Relation to the FCS programme

The framework remains complementary to FCS:

- **FCS at a point**: given specific integer weights, does the property hold? Lustre encoding plus Kind2 model checking, deterministic.
- **Population frameworks at a region**: across continuous weight space, where does the bifurcation lie? Rate-equation analysis with closed-form objects.
- **Closed-form thread (this note)**: the same rate equations but with $\Phi(\mu, \sigma)$, $H(\omega)$, and finite- N corrections derived from the spike physics. Inputs are physical (drive, noise variance, connectivity matrix); no hand-tuned sigmoid parameters.

A spectral pre-filter for Kind2 is the natural composition: certify a weight cell is non-reachable via Phase 1’s Siegert FP enumeration, and skip the model check. That is engineering follow-on; this note is prerequisite groundwork.

4.3. Limitations

The chief technical limitation surfaced by this thread is in τ_m **calibration** (Section 2.4): a static f - I fit constrains α and the gain at the operating point but does not constrain the **dynamic** time constant τ_m . Magnitude predictions in Phase 3 are biased by a factor of two as a result. Refitting τ_m on impulse-response or autocorrelation data would close this gap; we note it but do not implement it here.

The **quasi-renewal** implementation (Phase 4) uses the simplest mesoscopic correction (single-integral, memoryless hazard, $\tau_{\text{ref}} = 0$). Schwalger et al. 2017 [5] give the full age-structured integral with non-renewal corrections, which would sharpen the predicted boundary further. We declared full Schwalger out of scope at the planning stage and the Phase 4 result already passes the H3 gate; the upgrade is documented as follow-on.

The **Brunel–Hakim parabolic-cylinder** transfer function would replace the single-pole approximation in Equation 2 with the exact LI&F linear response. We use Richardson’s threshold-integration recipe [3] in its small- ω limit; the high-frequency resonance term is omitted by design. Phase 2’s self-consistency check verifies the omission is benign at $\omega = 0$; Phase 3’s residual phase error at high ω may partly trace to it.

5. Conclusion

Three closed-form objects – the Siegert formula [1], [2] (static), the Richardson single-pole transfer function [3] (dynamic), and the Naud–Gerstner quasi-renewal mesoscopic equation [4] (finite- N) – act on the same finite tangible topology and span the regimes of LI&F-archetype analysis. The static piece replaces the heuristic sigmoid with a physically-derived f - I curve depending on input mean **and** noise variance, beating the population-thread baseline on the contralateral WTA boundary (Jaccard 0.843 vs 0.796). The dynamic piece reduces the topology to a block diagram amenable to classical control theory, with self-consistency exact at $\omega = 0$ and phase-domain agreement within 10° on the FCS negative loop. The finite- N piece recovers the

spike-timing-locked rectangular bistability that the companion smooth-rate thread documented as a failure mode, with the predicted $1/\sqrt{N}$ approach to the mean-field Siegert wedge.

The framework is a **third leg** alongside the discrete-spectral analysis of `deq/archetypes/` and the smooth-rate analysis of `deq/population/`. Where `archetypes` verifies properties at points and `population` predicts shapes of regions, this thread predicts both shapes and dynamics with closed-form objects whose every parameter is a measurable LI&F quantity rather than a fitted curve coefficient. The topology stays explicit; the spike physics stays in.

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